

Om Patil

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EDUCATION

Sardar Patel Institute of Technology

B. Tech in Electronics and Telecommunications Engineering (CGPA: **9.11**)

Mumbai, IN

Nov 2022 – Present

SP Jain Institute of Management and Research

Minor in Finance and Management

Mumbai, IN

Jan 2024 – Present

EXPERIENCE

Intern

Jul 2025 – Aug 2025

GAIL (Gas Authority of India Limited)

- Worked with GAIL to further develop Samarth AI, a chatbot designed to enhance internal operations by enabling faster, organization-wide access to HR policies, technical manuals, and institutional documents.
- Developed an LLM-powered system that interprets user queries, extracts relevant tabular data from Excel and PDF files, and generating context-specific visual graphs to support data-driven insights.

Technical Assistant

Jan 2024 – Mar 2024

Sardar Patel Institute of Technology

- Developed a responsive webpage for the NAAC accreditation process, coordinated with college administration and IT department to fulfill institutional standards and functional requirements.

PROJECTS

Samarth AI | FastAPI, ChromaDB, Meta Llama

- Led the development of a **RAG-based chatbot** to assist employees in navigating HR and technical documents.
- Leveraged **contextual chunk headers** to enhance LLM relevance, raising top-1 precision by 34%.
- Built an **LLM-driven routing system** to map user intents to the correct task handlers within the agent.

Clover | C++, SDL3, GLM

- Engineered a 2D game engine in **C++** with **SDL**, using **OOP** to organize rendering, entities, and game state.
- Integrated a **fixed timestep update system** for consistent animation, physics, and logic across all systems.
- Constructed a **custom AABB collision system** handling **60+ objects** per frame with precise resolution.
- Implemented a centralized **resource manager** for managing game assets like textures, audio, and fonts, reducing redundant loading and improving overall responsiveness.

RFID Blood Bank Automation | C++, ESP32, MFRC522, Django

- Automated real-time blood tracking with **MFRC522 RFID** and **ESP32**, cutting manual logging time by 60%.
- Estimated prevention of **25+ expired units/month** through smart expiry alerts and inventory tracking.
- Automated donor certificate and notification system using email services to enhance outreach and engagement.

TECHNICAL SKILLS

Languages: C++, Python, SQL

Databases: MongoDB, PostgreSQL, ChromaDB, FAISS

Developer Tools: Git, Postman, VS Code, PyCharm, CLion, Arduino IDE

Libraries: Langchain, SDL3, GLM, Pandas, NumPy, Matplotlib

ACHIEVEMENTS

Smart India Hackathon 2024 Winner - Ranked 1st out of 166 teams in a prestigious national-level hackathon.

MathWorks IOT Competition 2024 Winner - Secured 1st place among 27 teams in a college level competition.

HERE India Hackathon 2025 Finalist – Placed in Top 10 among 40+ teams.

Codechef: 3-Star (Max Rating: **1628**)

EXTRACURRICULAR ACTIVITIES

Volunteer, Afroz Shah Foundation

Contributed 30+ volunteer hours to beach cleaning initiatives, actively removing plastic waste and participating in efforts to restore Mumbai's coastline and promote long-term environmental sustainability.